

Tutorial 2 Memo

Question 1

1. Using the data in Question 1 of Tutorial 1 (test_anxiety.csv), fit the model in R using the aov() function and extract the model estimates and ANOVA table. Do they match what you obtained with the manual calculations?

See Rscript. Do you know what each value represents? Feel free to check with your tutor or use the discussion board.

2. Based on the estimate of the standard error for differences in treatment means can you calculate the estimated value of σ^2 ? Hint: use the second formula shown in section 6.4 of the notebook.

$$Var(\hat{u}_i - \hat{u}_k) = Var(\hat{u}_i) + Var(\hat{u}_k) = \frac{\sigma^2}{n_i} + \frac{\sigma^2}{n_k}$$

$$n_i = n_k = r$$

$$Var(\hat{u}_i - \hat{u}_k) = 2 * \frac{\sigma^2}{r}$$

σ^2 is estimated by s^2 and from model output: se of treatment difference: 0.9866 and $r = 5$

$$0.9886^2 = 2 * \frac{s^2}{5}$$

$s^2 = (0.9886^2 / 2) * 5 = 2.443$ - this is the MSE you calculated earlier just using the rounded se of differences from the output hence the slight difference.

3. Using the simulated data in Question 2 of Tutorial 1 (remember to use the same seed), fit the model in R using the aov() function and extract the ANOVA table.

See Rscript. Do you know what each value represents? Feel free to check with your tutor or use the discussion board.

4. For both analyses, briefly interpret the results. Your answer should refer to the research question and explain what the results suggest in practical terms, not just whether the test was significant.

The amount of therapy received had a significant effect on students' test anxiety levels. At least one duration of therapy resulted in a different mean level of test anxiety at the end of treatment, suggesting that therapy duration influences anxiety outcomes ($F(2,12) = 5.18, p = 0.0239$).

Baking temperature had a significant effect on the average rise height of the bread. Loaves baked at different temperatures did not all rise to the same extent, indicating that temperature influences bread rise under the conditions of this experiment ($F(3,44) = 4.12, p = 0.0116$).

5. In a completely randomized design, 15 experimental units were used in each of three treatments. Complete the following analysis of variance. At a 0.05 level of significance, is there a significant difference between the treatments?

Source of variation	Sum of squares	Df	Mean square	F	p-value
Treatments	1200	2	600	42	<0.01
Error	600	42	14.285...		
Total	1800	44			

The p-value can be estimated by using the tables with df 2 and 40, the largest critical value is 5.18 associated with 0.01 probability of obtaining 5.81 or more extreme. The test statistic is much larger than 5.81, which means the p-value (probability of obtaining a test statistic of 42 or more extreme) is a lot smaller than 0.01. In R:

$Pf(42, 2, 42, \text{lower.tail} = F) = 9.559907e-11$

Question 2

Experiment Description

A researcher wants to determine if different study environments impact student performance. Three groups of students, each with 5 participants, were randomly assigned to different study environments:

- Group 1: Quiet Library
- Group 2: Coffee Shop
- Group 3: Home

At the end of a week, students took a standardized test (score out of 100). The scores recorded are as follows:

Coffee Shop	Home	Library
-------------	------	---------

70	82	78
74	85	80
72	79	75
69	81	82
73	80	77

The data is available in a csv file called: study_environment.csv

6. Identify the following based on the experiment description:

- Treatment factor: **Study Environment**
- Factor levels: **Coffee shop, home, library**
- Treatments: **same as levels**
- Response: **test scores**
- Experimental units: **Individual students**
- Number of replicates: **5 per treatment group**

7. What is the treatment structure and how has randomization been conducted?

The treatment structure is **single-factor with three levels**. Randomization is conducted by **randomly assigning students to one of the three study environments** without blocking.

8. Based on the treatment structure and randomisation procedure you have identified what type of design has been used here?

This is **single-factor completely randomised design**.

9. Write down the population model equation used for analysing data from this type of experiment. Be specific.

$$y_{ij} = \mu + A_i + \varepsilon_{ij}$$

where:

- y_{ij} = response variable (TAI score) where $i = 1, 2, 3$ and $j = 1, \dots, 5$
- μ = overall mean
- A_i = treatment effect for therapy duration
- ε_{ij} = random error term and $\varepsilon_{ij} \sim N(0, \sigma^2)$

10. Write out the separate model equations for each treatment.

- Coffee Shop ($i = 1$): $y_{1j} = \mu + A_1 + \varepsilon_{1j}$

- Home ($i = 2$): $y_{2j} = \mu + A_2 + \varepsilon_{2j}$

- Library ($i = 3$): $y_{3j} = \mu + A_3 + \varepsilon_{3j}$

Where each $\varepsilon_{ij} \sim N(0, \sigma^2)$

11. Now, assuming that the model assumptions have been checked, fit the model in R using the `aov()` function and extract the model estimates.

12. What is the estimated mean for the Library treatment? Interpret the mean in words.

The mean response for the Library treatment is 78.4. This is interpreted as the students who were assigned the Library treatment (i.e. studied in a library) scored on average 78.4% on the test.

13. What is the estimated effect of the Home treatment? Interpret the effect in words.

The effect of the home treatment was estimated as 4.267. In other words, the students who studied at home scored, on average, 4.267% higher than the overall mean (mean performance of all students).

14. Write out the fitted model equations.

- Group 1: $\hat{Y}_{\{1j\}} = 77.1333 - 5.533$

- Group 2: $\hat{Y}_{\{2j\}} = 77.1333 + 4.267$

- Group 3: $\hat{Y}_{\{3j\}} = 77.1333 + 1.267$

15. What is the 95% confidence interval for each of the treatment means? Interpret in words the confidence interval for one of the means.

95% CI: $\hat{\mu}_i \pm t_v^{\alpha/2} \times SE(\text{estimate})$

$t_{12}^{0.025} = 2.179$ (using tables or `r: qt(0.025, 12, lower.tail = FALSE)`)

Coffee shop:

$$71.6 \pm 2.179 \times 1.061$$

$$(69.29; 73.92)$$

Home:

$$81.4 \pm 2.179 \times 1.061$$

$$(79.09; 83.71)$$

Library:

$$78.4 \pm 2.179 \times 1.061$$

$$(76.09; 80.71)$$

Interpretation of the 95% confidence interval for coffee shop mean:

We are 95% confident that the true population mean performance for the Coffee Shop treatment lies between 69.29 and 73.92.

This means that if we were to repeat this experiment many times and compute a confidence interval each time, approximately 95% of those intervals would contain the true population mean.

16. Perform an ANOVA hypothesis in R using the summary() function. Make sure the data is in the correct format (i.e. long format) and that the treatment variable is correctly read in as a factor.

```

> summary(model)
              Df Sum Sq Mean Sq F value    Pr(>F)
`Study Environment`  2  252.1   126.07    22.38 8.93e-05 ***
Residuals           12   67.6     5.63
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

17. What is the total sums of squares?

$$SST = SSB + SSE = 252.1 + 67.6 = 319.7$$

18. What is the treatment MS and MSE?

$$\text{Treatment MS} = 126.07$$

$$\text{MSE} = 5.63$$

19. How was the F-value obtained? Interpret the F-value in words.

$$F = \text{MSB} / \text{MSE} = 126.07 / 5.63 = 22.38$$

The variability between groups (MSB) was roughly 22 times larger than the variability within groups (MSE).

20. What is the conclusion of the hypothesis based on the p-value? Give the p-value and interpret it in words.

The p-value is 8.93×10^{-5} which is the probability of observing the test statistic or more extreme under the null hypothesis. This p-value is extremely small, indicating that it is extremely unlikely that the observed test statistic of 22.38 belongs to the null distribution of $F_{2,12}$. We have strong evidence against the null hypothesis of equal means and conclude that there is evidence to suggest that the treatment did have an effect.

Part 2

An experiment has been conducted for four treatments. Complete the following analysis of variance table.

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F
Treatments	? 3	450	? 150	? 1.78
Error	? 16	? 1350	84.375	
Total	? 19	1800		

Use $\alpha = 0.05$ to test for any significant differences.

$F_{3,16}$ is the null distribution

p-value: $\text{pf}(1.78, 3, 16, \text{lower.tail}=\text{F}) = 0.19...$

critical value: $\text{qf}(0.05, 3, 16, \text{lower.tail}=\text{F}) = 3.23...$

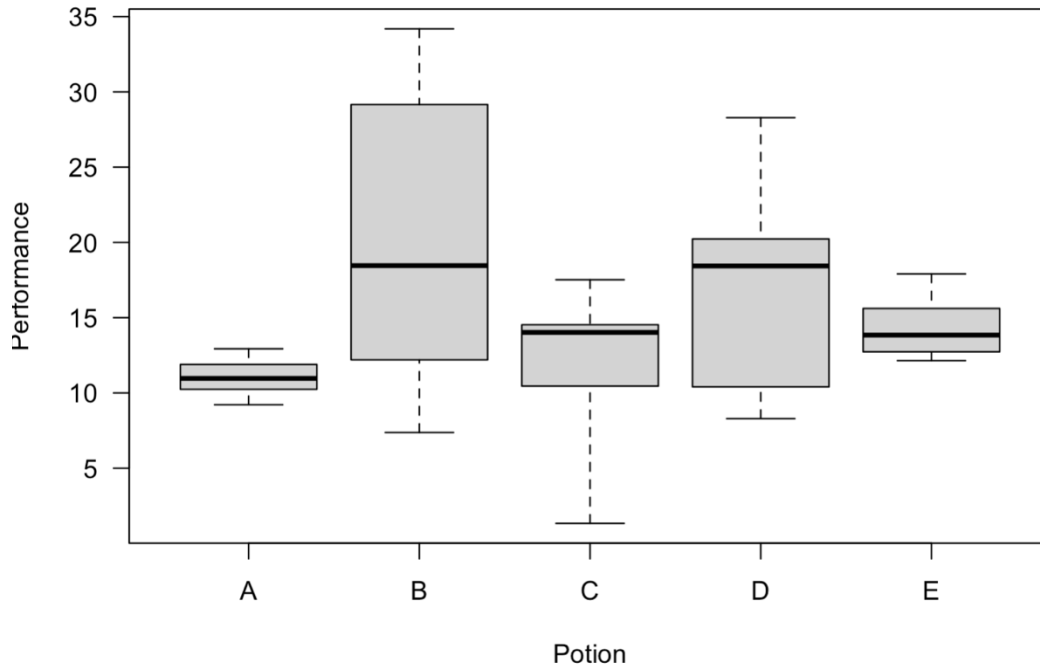
The p-value is large (>0.05) indicating that we do not have evidence against the null hypothesis of equal treatment means. We conclude that the treatment did not have an effect on the response.

The test statistic of 1.78 is smaller than the critical value indicating that we do not have evidence against the null hypothesis of equal treatment means. We conclude that the treatment did not have an effect on the response.

Part 3 – Assumption checking

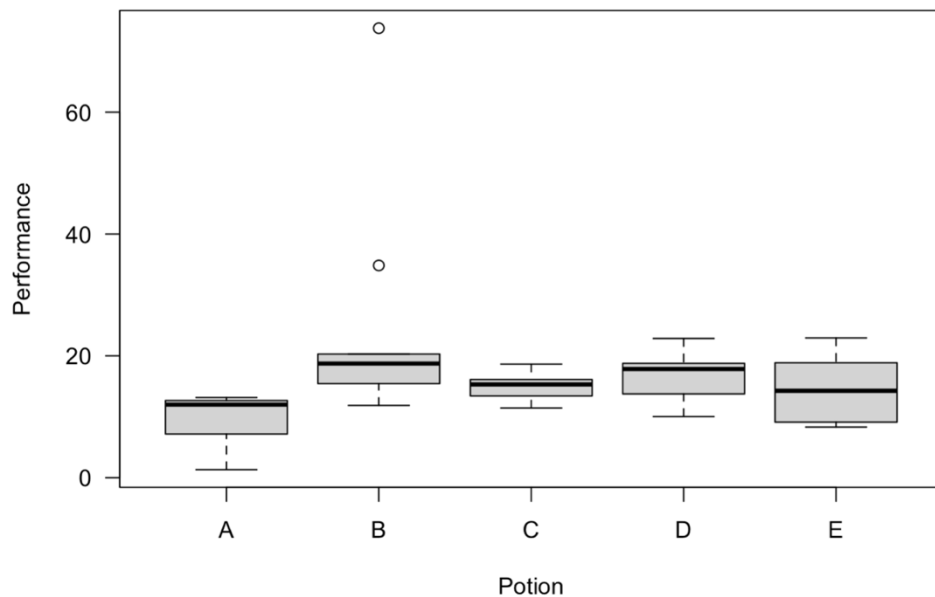
Identify in each of the plots below which assumption(s) is being checked and if you think there are any violations.

1.

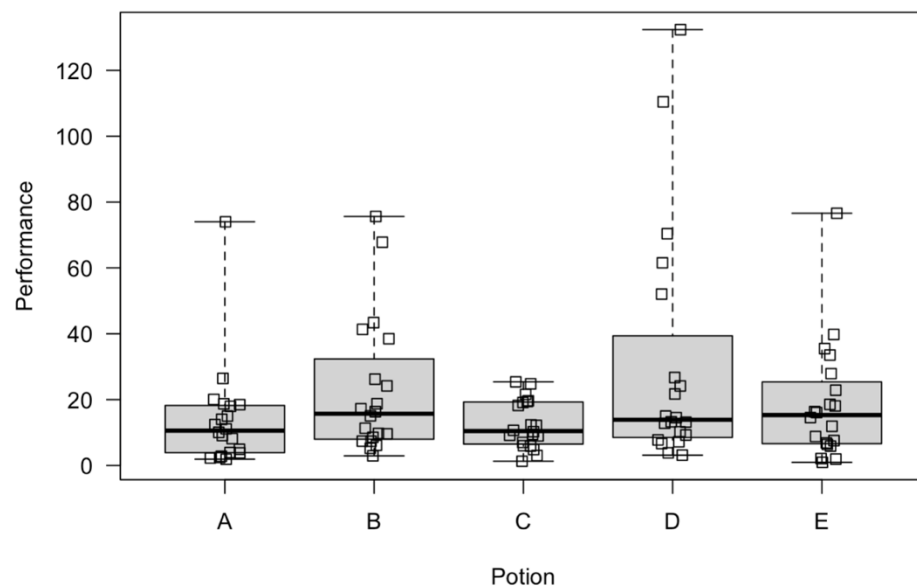


With boxplots we can check the assumption of no outliers, normality and equal population variance. There are no outliers, the boxplots look relatively symmetric but the heights (greyboxes) of the boxplots of quite different. The height of the second boxplot looks much larger than (probably more than 5 times larger) than the first one. The assumption of equal population variance seems like it is not met, we could also look at the sample standard deviations for a clearer picture.

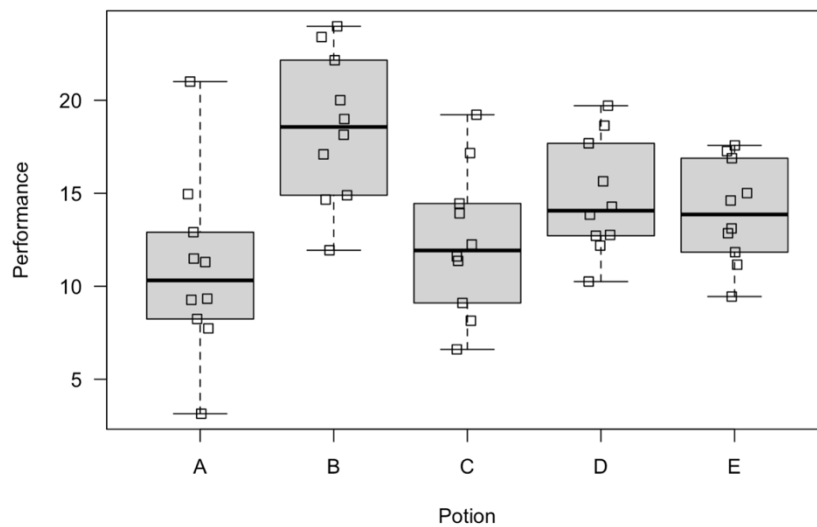
2. Same assumptions as in the previous one. Here there are clearly two outliers belonging to the second treatment.



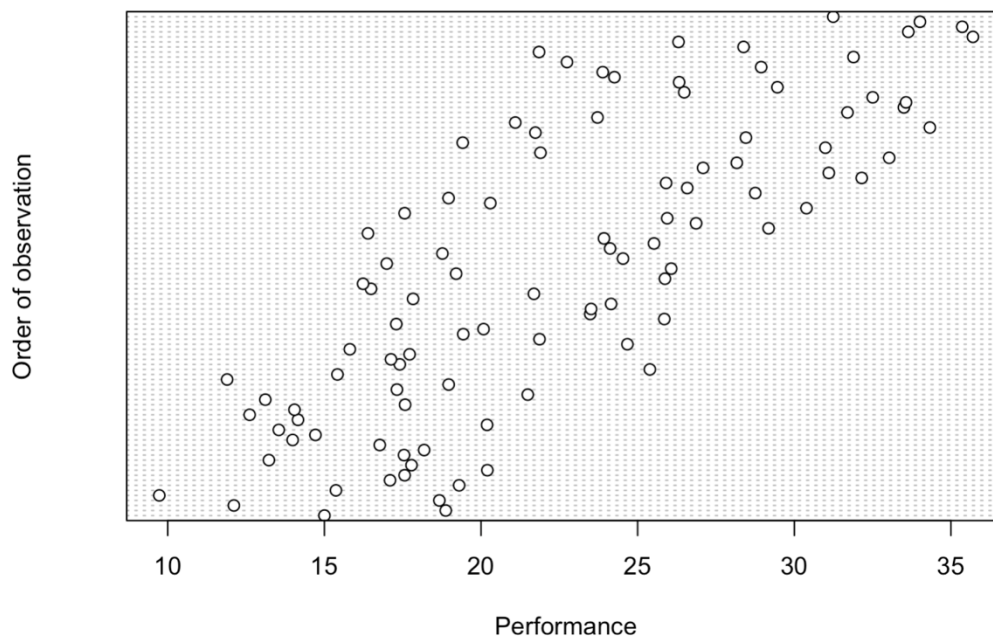
3. Same assumptions as in the previous one. Here all boxplots except that of treatment C shows skewed distributions. They seem to have long tails to the right suggesting that the observations do not follow a normal distribution.



4. There are no outliers, the boxplots are fairly symmetric and the heights are comparable. No assumptions which can be checked with the boxplots appears to be violated.



5. This plot shows the observations against the order in which they appear in the dataset. This is a dotchart used to check the assumption of dependence (specifically whether there is dependence between successive observations). The plot shows a clear trend in the observations. The points exhibit an upward trend, meaning that as the observation order increases, the performance values tend to increase. This suggests a strong dependence between successive observations. Ideally, in an independent dataset, the observations should appear randomly distributed without a clear trend.



Part 4 – RCBD

A taste-testing experiment has been designed so that four brands A to D of Colombian coffee are to be rated by nine experts. The experimenters are interested in the differences in mean ratings between the four brands. The tasting sequence of the four brands is randomly determined for each of the nine expert tasters until a rating on a 7-point scale (1 = extremely unpleasing, 7 = extremely pleasing) is given for each of four characteristics: taste, aroma, richness, and acidity. The data are the total rating, summed over all four characteristics.

Blocks (Experts)	A	B	C	D	Totals
E.B.	24	26	25	22	97
N.B.	27	27	26	24	104
M.D.	19	22	20	16	77
M.H.	24	27	25	23	99
B.J.	22	25	22	21	90
R.J.	26	27	24	24	101
B.K.	27	26	22	23	98
B.M.	25	27	24	21	97
J.S.	22	23	20	19	84
Totals	216	230	208	193	847

1. Identify the following based on the experiment description:
 - a. Treatment factor: **Coffee brand**
 - b. Factor levels: **A, B, C, D**
 - c. Treatments: **same as levels**
 - d. Response: **Total rating**
 - e. Experimental units: **This is a bit tricky, but we could see this as each individual rating given by an expert to a specific coffee brand is an experimental unit.**
 - f. Number of replicates: **9 replicates per treatment**

2. Why are experts blocks?

We are primarily interested in the differences between coffee brands. We are using 9 different experts, and they are treated as blocks because they introduce a potential source of variability that is not of primary interest but could influence the results. For example, some experts might consistently give higher ratings, while other might be more critical. To account for these differences, we block for experts and randomized treatments “within”.

3. What is the treatment structure and how has randomisation been conducted?

The treatment structure is **single-factor with four levels**. Randomization is conducted within blocks, i.e. to experts.

4. Based on the treatment structure and randomisation procedure you have identified what type of design has been used here?

This is single-factor RCBD.

5. Write down the population model equation used for analysing data from this type of experiment. Be specific.

$$Y_{ij} = \mu + A_i + B_j + \varepsilon_{ij}$$

where:

- Y_{ij} is the observed total rating for coffee brand i given by expert j
- μ is the overall mean rating
- A_i is the treatment effect of coffee brand i
- B_j is the block effect of expert j
- ε_{ij} = random error term and $\varepsilon_{ij} \sim N(0, \sigma^2)$

6. Calculate the following by hand and note down the R code you could use to calculate the first three:
- a. Overall mean

$$\bar{Y} = \frac{\sum Y_{ij}}{N}$$

$$\bar{Y} = \frac{847}{36} = 23.52778$$

- b. Treatment means and note down the formula for calculating their standard errors.

$$\bar{Y}_i = \frac{\sum_j Y_{ij}}{b}$$

$$\bar{Y}_A = \frac{216}{9} = 24.000$$

$$\bar{Y}_B = \frac{230}{9} = 25.5556$$

$$\bar{Y}_C = \frac{208}{9} = 23.1111$$

$$\bar{Y}_D = \frac{193}{9} = 21.4444$$

$$SE_{\bar{Y}_i} = \sqrt{\frac{MSE}{b}}$$

- c. Treatment effects and note down the formula for calculating their standard errors

$$\widehat{A}_A = 24.000 - 23.5278 = 0.4722$$

$$\widehat{A}_B = 25.5556 - 23.5278 = 2.0278$$

$$\widehat{A}_C = 23.1111 - 23.5278 = -0.4167$$

$$\widehat{A}_D = 21.4444 - 23.5278 = -2.0833$$

$$SE_{\widehat{A}_i} = \sqrt{\frac{MSE}{b}}$$

- d. Block means and effects

$$\bar{Y}_j = \frac{\sum_i Y_{ij}}{a}$$

$$\text{EB: } \bar{Y}_1 = \frac{97}{4} = 24.25$$

$$\text{NB: } \bar{Y}_2 = \frac{104}{4} = 26.00$$

$$\text{MD: } \bar{Y}_3 = \frac{77}{4} = 19.25$$

$$\text{MH: } \bar{Y}_4 = \frac{99}{4} = 24.75$$

Repeat for blocks 5 to 9

Block effects:

EB:

$$\widehat{B}_1 = 24.25 - 23.5278 = 0.722$$

$$\text{NB: } \widehat{B}_2 = 26.00 - 23.5278 = 2.472$$

$$\text{MD } \widehat{B}_3 = 19.25 - 23.5278 = -4.278$$

$$\text{MH: } \widehat{B}_4 = 24.75 - 23.5278 = 1.222$$

Repeat for blocks 5 to 9

7. What are the assumptions of this model?

1. No severe outliers
2. Normality: Errors are normally distributed.
3. Independence: Observations are independent within and across blocks.
4. Homogeneity of Variance
5. Additivity: No interaction between blocks and treatments.

8. What does the assumption of additivity between blocks and treatments mean? Put in context of the experiment.

Additivity assumes that the effect of coffee brand is consistent across all experts. If an interaction exists (i.e., some experts systematically prefer one brand more than others), the assumption is violated.

9. Assuming you have checked the assumptions and they are reasonably met. Conduct the ANOVA hypothesis by hand and summarise the results in an ANOVA table.

Calculations are done using unrounded values, rounded values are shown for brevity.

SSTr (SS Treatment)

$$SSTr = b \sum (\bar{Y}_t - \bar{Y})^2$$

$$SSTr = 9 \times [(0.4722)^2 + (2.0278)^2 + (-0.4167)^2 + (-2.0833)^2]$$

$$SSTr = 9 \times 8.848765 = 79.63889$$

SSB (SS Blocks)

$$SSB = a \sum (\bar{Y}_j - \bar{Y})^2$$

$$SSB = 4 \times [(23.00 - 23.5278)^2 + (26.00 - 23.5278)^2 + \dots + (21.00 - 23.5278)^2]$$

$$SSB = 4 \times 38.30556$$

$$SSB = 4 \times 38.2314 = 153.22222$$

SST (SS Total)

$$SST = \sum Y_{ij}^2 - \frac{(\sum Y_{ij})^2}{N}$$

$$SST = 5240 + 5906 + 4846 + 4193 - \frac{847^2}{36}$$

$$SST = 256.9722$$

SSE

$$SSE = SST - SSTr - SSB$$

$$SSE = 256.9722 - 153.22222 - 79.63889 = 24.11109$$

Mean Squares

$$MSTr = \frac{SSTr}{t - 1} = \frac{79.63889}{3} = 26.5463$$

$$MSB = \frac{SSB}{b - 1} = \frac{153.22222}{8} = 19.1528$$

$$MSE = \frac{SSE}{(t - 1)(b - 1)} = \frac{24.11109}{(4 - 1)(9 - 1)} = \frac{24.11109}{24} = 1.0046$$

Response: Score

	Df	Sum Sq	Mean Sq	F value
Experts	8	153.222	19.1528	19.064
Category	3	79.639	26.5463	26.424
Residuals	24	24.111	1.0046	

10. What conclusion can you draw about the treatment factor based on the result you obtained? Provide a p-value to support your conclusion.

There is strong evidence against the null hypothesis of equal means as indicated by the extremely small p-value. This suggests that at least one mean rating of the coffee brands is different.

Using tables $p < 0.01$

Using R: 8.863×10^{-8} . The correct code is: `pf(26.424, 3, 24, lower.tail = FALSE)`.

11. Was blocking effective in this experiment? Motivate your answer.

The blocking variable accounted for approximately 60% of the total sum of squares in the response leading to an F-value greater than one for the blocking factor. This indicates that blocking was effective by reducing the experimental error variance.